

The kinetic-relation stability criterion for traveling kinks in damped, driven lattices: a Jordan-chain identity

Research note — draft for discussion

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Abstract

In a damped, dc-driven Frenkel–Kontorova lattice, Vainchtein, Cuevas-Maraver, Kevrekidis and Xu found numerically that traveling kinks acquire an unstable direction exactly where the kinetic relation $\sigma(c)$ between driving force and velocity has an extremum, and identified the obstruction to a proof: dissipation destroys the generalized eigenvector at the neutral Floquet multiplier on which the Hamiltonian argument rests [1]. We show that this obstruction can be measured exactly. Writing the co-traveling linearization as a quadratic pencil $Q(\nu) = \nu^2 + \nu M_1 + M_0$, we prove the operator duality $Q(-\gamma - \nu) = Q(\nu)^\top$, which contains the conformally symplectic structure of the monodromy and identifies the left neutral eigenvector with the null vector $\hat{p}$ of the *anti-damped* advance–delay equation. Under hypotheses on $\hat{p}$ that we verify numerically but do not prove, the scalar $\kappa(c) = \langle \hat{p}, M_1 \phi' \rangle$ that decides whether the neutral eigenvalue is simple satisfies $\kappa(c) = -\mu \sigma'(c) \langle \hat{p}, \mathbf{1} \rangle$, and, along the branch parametrized by arclength, $\dot{c} \kappa + \mu \dot{\sigma} \langle \hat{p}, \mathbf{1} \rangle = 0$. Hence a multiplier can reach +1 only at an extremum of σ , while at a turning point of the branch in c it is $\langle \hat{p}, \mathbf{1} \rangle$ that vanishes and no crossing occurs. A Lyapunov–Schmidt reduction then reduces the local sufficiency of the slope criterion to the sign of one computable coefficient. A Fourier-spectral solver for the advance–delay equation, independent of the method of [1], reproduces its critical velocity, critical force and maximal velocity, verifies the identity to 10^{-13} with spectral convergence, and confirms the predicted behaviour at the first extremum and the first turning point. The identity extends to damped, additively driven lattices with arbitrary periodic onsite potential and symmetric finite-range coupling. We state precisely what is proved, what is numerical, and what remains open.

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1 Introduction

1.1 The problem

We consider the damped, dc-driven Frenkel–Kontorova (FK) lattice

$$\ddot{u}_n + \gamma \dot{u}_n = u_{n+1} - 2u_n + u_{n-1} + \mu(\sigma - \sin u_n), \quad n \in \mathbb{Z}, \quad (1)$$

with damping $\gamma > 0$, onsite strength $\mu > 0$ and driving force $\sigma \in (0, 1)$, and its traveling kinks $u_n(t) = \phi(\xi)$, $\xi = n - ct$, which satisfy the advance–delay equation

$$c^2 \phi''(\xi) - \gamma c \phi'(\xi) = \phi(\xi + 1) - 2\phi(\xi) + \phi(\xi - 1) + \mu(\sigma - \sin \phi(\xi)), \quad (2)$$

$$\phi(\xi) \rightarrow \arcsin \sigma + 2\pi \quad (\xi \rightarrow -\infty), \quad \phi(\xi) \rightarrow \arcsin \sigma \quad (\xi \rightarrow +\infty), \quad \phi(0) = \pi. \quad (3)$$

For $\gamma > 0$ the traveling waves come in one-parameter families along which the driving force is a function of the velocity, the *kinetic relation* $\sigma = \sigma(c)$. This relation is the central constitutive object in the continuum description of lattice defects (dislocations, fronts, domain walls) and has been computed repeatedly, together with the observation that its decreasing portions are unstable [7, 8, 9, 10, 6].

Vainchtein, Cuevas-Maraver, Kevrekidis and Xu [1] computed the traveling kinks of (1) by Newton iteration on the period map of the lattice, determined their Floquet multipliers, and found

a precise numerical criterion: portions of the kinetic curve with $\sigma'(c) < 0$ always carry an unstable direction, a real multiplier leaving the unit circle through +1 exactly at the extrema of σ , and a further unstable direction appears at every subsequent extremum of the spiralling, multivalued kinetic curve. They note that the sharpness of the criterion makes it “highly likely to be associated with a theorem”, and report that the Hamiltonian argument [5, 2, 3] fails for two reasons. The first is technical: the waves decay slowly to a nontrivial asymptotic state, so the function spaces and inner products must be adapted. The second is conceptual: in the Hamiltonian problem the multiplier $\rho = 1$ carries both the translation eigenvector $\partial_\xi \phi$ and the generalized eigenvector $\partial_c \phi$, and the criterion follows from symplectic orthogonality against the latter; with damping this generalized eigendirection is eliminated, its multiplier moving to $e^{-\gamma/c}$. They suggest that a suitably weighted space might restore the Hamiltonian frame.

1.2 Summary of results

The present note quantifies the second obstruction exactly and locates the first. Its results, with their status, are as follows.

- (i) *Structure of the linearization* (§3, proved). The monodromy M of the linearized lattice is conformally symplectic, $M^\top J M = e^{-\gamma/c} J$. The co-traveling pencil satisfies the stronger operator identity $Q(-\gamma - \nu) = Q(\nu)^\top$, which gives the multiplier pairing $\rho \hat{\rho} = e^{-\gamma/c}$, places the essential spectrum on $\text{Re } \nu = -\gamma/2$, and identifies the left neutral eigenvector with the null vector $\hat{p}$ of the anti-damped equation ($\gamma \rightarrow -\gamma$). The weighted space suggested in [1] exists in the form of the scalar rescaling $e^{\gamma/(2c)} M$, which is exactly symplectic, but it moves the stability boundary off the unit circle and does not by itself restore the Hamiltonian argument (Remark 3.5).

- (ii) *The Jordan-chain identity* (§4, proved under Hypotheses H1–H3). The scalar $\kappa(c) = \langle \hat{p}, M_1 \phi' \rangle$, $M_1 = -2c \partial_\xi + \gamma$, whose vanishing is equivalent to the existence of a generalized eigenvector at $\nu = 0$, satisfies

$$\kappa(c) = -\mu \sigma'(c) \langle \hat{p}, \mathbf{1} \rangle, \quad \langle \hat{p}, \cos \phi \rangle = 0. \quad (4)$$

With the normalization $\langle \hat{p}, \mathbf{1} \rangle = \langle \phi', \mathbf{1} \rangle = -2\pi$ this reads $\kappa = 2\pi \mu \sigma'(c)$: the generalized eigendirection is destroyed by the damping by exactly $2\pi \mu \sigma'(c)$, and is restored, as $-\partial_c \phi$, precisely where $\sigma'(c) = 0$.

- (iii) *Turning points* (§5, proved under Hypotheses H1–H3). Along the branch parametrized by arclength, $\dot{c} \kappa + \mu \dot{\sigma} \langle \hat{p}, \mathbf{1} \rangle = 0$. At an extremum of σ this forces $\kappa = 0$; at a turning point in c it forces $\langle \hat{p}, \mathbf{1} \rangle = 0$ instead. The two kinds of critical point on the multivalued kinetic curve are thus distinguished by which factor of (4) vanishes.
- (iv) *Local crossing* (§6, proved under Hypotheses H1–H3 and three non-degeneracy conditions). Near an extremum $\hat{c}$, the pencil has exactly two eigenvalues near 0: $\nu = 0$ and a real $\nu_2(c)$ with $\nu_2 \approx -\kappa/m$ and $\nu_2'(\hat{c}) = \mu \sigma''(\hat{c}) \langle \hat{p}, \mathbf{1} \rangle / m(\hat{c})$, where m is an explicit coefficient whose vanishing at $\hat{c}$ is equivalent to a Jordan chain of length three. The slope criterion holds near $\hat{c}$ if and only if $m(\hat{c})$ has the sign fixed in Corollary 6.2.
- (v) *Power balance* (§4, proved for decaying waves). $\mu \sigma = (\gamma c / 2\pi) \int_{\mathbb{R}} \phi'(\xi)^2 d\xi$.
- (vi) *Generality* (§7, proved under the analogous hypotheses). The identity holds for damped lattices with additive drive f , any smooth 2π -periodic onsite potential and any symmetric finite-range coupling: $\kappa = -f'(c) \langle \hat{p}, \mathbf{1} \rangle$.

- (vii) *Numerics* (§8–9, numerical). A Fourier-spectral solver for (2) reproduces $\hat{c}_1$, $\sigma(\hat{c}_1)$ and $c_{\max}$ of [1]; confirms (4) to 10^{-13} with spectral convergence and independence of the domain size; confirms that the stability transition and the zero of σ' coincide to 10^{-6} in c ; confirms the turning-point dichotomy at the first fold; finds $\hat{p}$ exponentially localized; and confirms the generalized identity in four modified lattices.

What is *not* established is listed in §10. The main gaps are the functional-analytic Hypotheses H2–H3 (simplicity of the kernel and exponential decay of $\hat{p}$ on the infinite lattice; the Fredholm property itself is proved in Lemma 2.1), the sign of m , any statement beyond a neighbourhood of each extremum, crossings through $\rho = -1$ or off the real axis, and the weakly damped resonance regime.

1.3 Relation to other stability criteria

For Hamiltonian lattices the analogous criterion is the change of monotonicity of the energy along the family, for traveling waves [5, 2, 3] and for discrete breathers [4]. In those settings $\partial_c \phi$ is a generalized eigenvector for every c and the criterion lives one Jordan level higher. The present structure is also dual to that of the Vakhitov–Kolokolov criterion for solitons of non-Hermitian nonlinear Schrödinger equations [19]: there the $U(1)$ Jordan block survives any gain–loss profile and the relevant scalar is one level up, whereas here the damping destroys translation’s partner and the scalar sits at the first level.

1.4 Status conventions

Every statement below carries one of the tags **proved**, **proved under ...**, **numerical** or **open**. A numerical statement is supported by convergence-checked computation but not proved; a statement proved under hypotheses is not claimed beyond them.

2 Setting

2.1 Linearization, monodromy and the co-traveling pencil

Linearizing (1) about $u_n(t) = \phi(n - ct)$ gives

$$\ddot{\xi}_n + \gamma \dot{\xi}_n = \xi_{n+1} - 2\xi_n + \xi_{n-1} - \mu \cos(u_n(t)) \xi_n. \quad (5)$$

With $z = (\xi, \eta)$, $\eta = \dot{\xi}$, this is $\dot{z} = A(t)z$,

$$A(t) = \begin{pmatrix} 0 & I \\ K(t) & -\gamma I \end{pmatrix}, \quad K(t) = \Delta - \mu \operatorname{diag}(\cos u_n(t)) = K(t)^\top, \quad (6)$$

with Δ the lattice Laplacian. Following [2, 1], the traveling wave is a fixed point of “evolve for $T = 1/c$, then shift by one site”, and its spectral stability is governed by the monodromy

$$M = S \Phi(T), \quad (Sz)_n = z_{n+1}, \quad (7)$$

where Φ is the fundamental matrix of $\dot{z} = A(t)z$. On a finite lattice with the charge-preserving boundary conditions of [1], the perturbation ξ is cyclic and S is a permutation.

Seeking Floquet solutions $\xi_n(t) = e^{\nu t} p(n - ct)$ of (5) gives the quadratic eigenvalue problem (Appendix A)

$$Q(\nu)p := (\nu^2 + \nu M_1 + M_0)p = 0, \quad \rho = e^{\nu/c}, \quad (8)$$

$$M_0 = c^2 \partial_\xi^2 - \gamma c \partial_\xi - \Delta_1 + \mu \cos \phi, \quad M_1 = -2c \partial_\xi + \gamma, \quad \Delta_1 p = p(\cdot + 1) + p(\cdot - 1) - 2p. \quad (9)$$

M_0 is the linearization of (2) at fixed (σ, c) . The map $\nu \mapsto \rho$ is $2\pi ic$ -periodic; near the real axis this plays no role. Throughout, instability means an eigenvalue with $\operatorname{Re} \nu > 0$, i.e. a multiplier with $|\rho| > 1$; we rely on [2, 3] for the correspondence between the pencil and the monodromy formulations.

2.2 Pairing, transpose, hypotheses

We use the *bilinear* pairing $\langle f, g \rangle = \int_{\mathbb{R}} f(\xi)g(\xi) d\xi$ and write $T^\top$ for the formal transpose with respect to it. Since $\partial_\xi^\top = -\partial_\xi$ and $\Delta_1^\top = \Delta_1$,

$$M_0^\top = c^2 \partial_\xi^2 + \gamma c \partial_\xi - \Delta_1 + \mu \cos \phi, \quad M_1^\top = 2c \partial_\xi + \gamma, \quad (10)$$

i.e. $M_0^\top$ is M_0 with $\gamma \rightarrow -\gamma$. Let C_b^2 denote C^2 functions that are bounded together with their first two derivatives. The following hypotheses are used below.

Hypothesis H1 (Branch). On an interval I of velocities (or, in §5, along a C^1 curve in arclength s) there is a C^1 family (ϕ, σ) of solutions of (2)–(3) such that ϕ' , ϕ'' and $\phi - \phi_\pm$ decay exponentially as $\xi \rightarrow \pm\infty$ ($\phi_\pm$ the asymptotic states), the derivative of the family with respect to the parameter lies in C_b^2 , and $\phi'(0) \neq 0$.

Hypothesis H2 (Simple kernel). For each parameter value, $\ker M_0 = \operatorname{span}\{\phi'\}$ in $H^2(\mathbb{R})$.

Hypothesis H3 (Adjoint decay). The null vector $\hat{p}$ of $M_0^\top$ (unique up to scaling under H2, by Lemma 2.1 below) is C^2 , and $\hat{p}, \hat{p}', \hat{p}''$ decay exponentially. In particular $\hat{p} \in L^1$.

The Fredholm property itself need not be assumed.

Lemma 2.1 (Fredholm property; proved under H1). *Let $\gamma > 0$ and $\mu\sqrt{1-\sigma^2} > \gamma^2/4$. For every ν with $\operatorname{Re} \nu > -\gamma/2$, $Q(\nu) : H^2(\mathbb{R}) \rightarrow L^2(\mathbb{R})$ is Fredholm of index zero, and $\nu \mapsto Q(\nu)$ is analytic there. In particular $M_0 = Q(0)$ is Fredholm of index zero; if H2 holds, $\ker M_0^\top$ is spanned by a single $\hat{p} \in H^2$ and $\operatorname{Ran} M_0 = \{f \in L^2 : \langle \hat{p}, f \rangle = 0\}$.*

Proof. Let $Q_\infty(\nu)$ be $Q(\nu)$ with $\cos \phi$ replaced by its common limit $\sqrt{1-\sigma^2}$ at $\pm\infty$. It is a Fourier multiplier with symbol $s(\nu, k) = \nu^2 + \nu(\gamma - 2ick) + L(k)$, where $L(k) = -c^2 k^2 - i\gamma ck + 4\sin^2(k/2) + \mu\sqrt{1-\sigma^2}$ (cf. [1, eq. (7)]). Writing $\nu = \lambda + ick$ gives $s = \lambda^2 + \gamma\lambda + \omega_k^2$ with $\omega_k^2 = 4\sin^2(k/2) + \mu\sqrt{1-\sigma^2} > \gamma^2/4$, so $s(\nu, k) = 0$ for real k only if $\operatorname{Re} \nu = -\gamma/2$. For $\operatorname{Re} \nu > -\gamma/2$ the symbol is continuous, nonvanishing and $\sim -c^2 k^2$ as $|k| \rightarrow \infty$, hence $|s(\nu, k)| \geq C(1+k^2)$ and $Q_\infty(\nu) : H^2 \rightarrow L^2$ is an isomorphism. The difference $Q(\nu) - Q_\infty(\nu) = \mu(\cos \phi - \sqrt{1-\sigma^2})$ is multiplication by a bounded continuous function tending to zero at $\pm\infty$ by H1, which is compact from H^2 to L^2 . Fredholmness with index zero follows from stability of the index under compact perturbations. Since the coefficients are real, the L^2 -adjoint has the same kernel as the transpose; bootstrapping in $M_0^\top \hat{p} = 0$ gives $\hat{p} \in H^2$. $\square$

Remark 2.2 (Status of the hypotheses). Lemma 2.1 settles the Fredholm question. What remains is H2, simplicity of the kernel, which is not automatic, and H3, exponential decay of $\hat{p}$ beyond the L^2 membership supplied by Lemma 2.1. Since the limiting operator is hyperbolic ($L(k) \neq 0$ for real k), exponential decay of H^2 kernel elements is expected from the theory of functional differential equations of mixed type [12]; we have not carried this out (open). H1 is not addressed analytically. All three hypotheses are supported numerically (§9).

3 Structure of the linearized problem

Proposition 3.1 (Conformal symplecticity; proved). *Let $J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$. For any symmetric $K(t)$ in (6), $A^\top J + JA = -\gamma J$. Consequently $\Phi(t)^\top J \Phi(t) = e^{-\gamma t} J$ and, since S is symplectic,*

$$M^\top J M = e^{-\gamma/c} J. \quad (11)$$

Proof. Direct multiplication gives $A^\top J = \begin{pmatrix} -K & 0 \\ \gamma I & I \end{pmatrix}$ and $JA = \begin{pmatrix} K & -\gamma I \\ 0 & -I \end{pmatrix}$, whose sum is $\begin{pmatrix} 0 & -\gamma I \\ \gamma I & 0 \end{pmatrix} = -\gamma J$. Hence $\frac{d}{dt}(\Phi^\top J \Phi) = \Phi^\top (A^\top J + JA) \Phi = -\gamma \Phi^\top J \Phi$ with $\Phi(0) = I$. Finally $S = \text{diag}(s, s)$ with $s^\top s = I$, so $S^\top J S = J$. $\square$

Only three properties of (1) enter: Newtonian second-order form, a symmetric force Jacobian, and uniform scalar damping. Conformally symplectic structure of damped Hamiltonian flows is classical [13]; what matters here is its consequence for the neutral direction.

Corollary 3.2 (Multiplier pairing and left eigenvectors; proved). *The spectrum of M is invariant under $\rho \mapsto e^{-\gamma/c}/\rho$. If $Mv = \rho v$ then $M^\top(Jv) = (e^{-\gamma/c}/\rho)(Jv)$: the left eigenvector at $e^{-\gamma/c}/\rho$ is Jv . In particular the multiplier $\rho = 1$ of translation is paired with $\hat{\rho} = e^{-\gamma/c}$, recovering the structure reported in [1, 8].*

The pencil carries the same information in a form that exhibits the adjoint problem directly.

Proposition 3.3 (Pencil duality; proved). *As formal operators on the line, $Q(-\gamma - \nu) = Q(\nu)^\top$ for all $\nu \in \mathbb{C}$.*

Proof. Using (9),

$$\begin{aligned} Q(-\gamma - \nu) &= (\gamma + \nu)^2 + (-\gamma - \nu)(-2c\partial_\xi + \gamma) + M_0 \\ &= \nu^2 + \nu(2c\partial_\xi + \gamma) + [c^2\partial_\xi^2 + \gamma c\partial_\xi - \Delta_1 + \mu \cos \phi] = \nu^2 + \nu M_1^\top + M_0^\top, \end{aligned}$$

where the γ^2 terms cancel and $2\gamma c\partial_\xi - \gamma c\partial_\xi = \gamma c\partial_\xi$. By (10) the right-hand side is $Q(\nu)^\top$. $\square$

Corollary 3.4 (proved). *(a) The spectrum of the pencil is invariant under $\nu \mapsto -\gamma - \nu$, i.e. $\rho\hat{\rho} = e^{-\gamma/c}$. (b) The left null vector of $Q(\nu)$ is the right null vector of $Q(-\gamma - \nu)$. In particular the left null vector at $\nu = 0$ solves $M_0^\top \hat{p} = 0$, the anti-damped version of the linearized traveling-wave equation; equivalently, $\hat{p}$ generates the eigenvalue $\nu = -\gamma$, $\hat{\rho} = e^{-\gamma/c}$. (c) For the constant-coefficient pencil obtained by replacing $\cos \phi$ by its limit $\sqrt{1 - \sigma^2}$, every spectral value satisfies $\text{Re } \nu = -\gamma/2$, provided $\mu\sqrt{1 - \sigma^2} > \gamma^2/4$.*

Proof. (a) and (b) are immediate from Proposition 3.3. For (c), plane waves $\xi_n = e^{\lambda t + iqn}$ of (5) with constant coefficients satisfy $\lambda^2 + \gamma\lambda + \omega_q^2 = 0$, $\omega_q^2 = 4\sin^2(q/2) + \mu\sqrt{1 - \sigma^2}$, so $\text{Re } \lambda = -\gamma/2$ whenever $\omega_q^2 > \gamma^2/4$, which holds for all q under the stated condition; and $\nu = \lambda + icq$ for $p(\xi) = e^{iq\xi}$. $\square$

By Lemma 2.1, the pencil is Fredholm of index zero to the right of this line, so $|\rho| = e^{-\gamma/(2c)}$ bounds the “multiplier circle” of [1] from outside; that the whole line belongs to the essential spectrum follows from standard Weyl-sequence arguments, which we do not repeat. For the parameters used below ($\mu = 1$, $\gamma = 0.1$, $\sigma \approx 0.65$), $\mu\sqrt{1 - \sigma^2} \approx 0.76 \gg \gamma^2/4$.

Remark 3.5 (Two candidate weights; proved). (1) By (11), $\tilde{M} = e^{\gamma/(2c)}M$ satisfies $\tilde{M}^\top J \tilde{M} = J$: a weight restoring symplecticity exists and is a scalar. It maps the stability boundary $|\rho| = 1$ to $|\tilde{\rho}| = e^{\gamma/(2c)} \neq 1$, and the translation pair $(1, e^{-\gamma/c})$ to the hyperbolic pair $(e^{\gamma/(2c)}, e^{-\gamma/(2c)})$. The Jordan block at $\rho = 1$ used in the Hamiltonian proof is therefore unfolded by an amount depending only on γ/c , and the rescaling does not by itself reduce the stability question to a Hamiltonian one. (2) The substitution $\chi = e^{-\gamma\xi/c}\psi$ intertwines the *differential* parts of M_0 and $M_0^\top$ exactly, but maps the shifts to $e^{\mp\gamma/c}\psi(\xi \pm 1)$, so it does not intertwine M_0 with $M_0^\top$. Numerically, the tails of $\hat{p}$ mirror those of ϕ' rather than carrying this weight (Observation 9.5). The route taken below bypasses weights altogether and uses Proposition 3.3.

4 The Jordan-chain identity

Lemma 4.1 (Pairing lemma; proved under H3). *If $M_0^\top \hat{p} = 0$ and $\hat{p}$ satisfies H3, then $\langle \hat{p}, M_0 v \rangle = 0$ for every $v \in C_b^2$.*

Proof. Integrating by parts, $\int \hat{p} c^2 v'' = \int c^2 \hat{p}'' v + c^2 [\hat{p} v' - \hat{p}' v]_{-\infty}^{\infty}$ and $\int \hat{p}(-\gamma c v') = \int \gamma c \hat{p}' v - \gamma c [\hat{p} v]_{-\infty}^{\infty}$; the boundary terms vanish because $\hat{p}, \hat{p}' \rightarrow 0$ and v, v' are bounded. The shifts satisfy $\int \hat{p}(\xi) v(\xi \pm 1) d\xi = \int \hat{p}(\xi \mp 1) v(\xi) d\xi$, all integrals converging absolutely since $\hat{p}, \hat{p}', \hat{p}'' \in L^1$ and $v, v', v'' \in L^\infty$. Hence $\langle \hat{p}, M_0 v \rangle = \langle M_0^\top \hat{p}, v \rangle = 0$. $\square$

The point of Lemma 4.1 is that the correct pairing is L^1 against L^∞ rather than L^2 against L^2 : this is what allows pairing against $\partial_c \phi$, which does not decay, and against the constant function. It is the concrete form of the “adapted spaces” called for in [1].

Lemma 4.2 (proved under H3). $\langle \hat{p}, \cos \phi \rangle = 0$.

Proof. $M_0 \mathbf{1} = \mu \cos \phi$ (derivatives and Δ_1 annihilate constants); apply Lemma 4.1 with $v = \mathbf{1}$. $\square$

Proposition 4.3 (Power balance; proved under H1). *Every solution of (2)–(3) with exponentially decaying ϕ' satisfies*

$$\mu \sigma = \frac{\gamma c}{2\pi} \int_{\mathbb{R}} \phi'(\xi)^2 d\xi. \quad (12)$$

Proof. Multiply (2) by ϕ' and integrate over $\mathbb{R}$. The term $c^2 \phi'' \phi'$ is an exact derivative. For the lattice term, $\int [\phi(\xi + 1) + \phi(\xi - 1)] \phi'(\xi) d\xi = \int \frac{d}{d\xi} [\phi(\xi) \phi(\xi - 1)] d\xi = \phi_+^2 - \phi_-^2 = \int 2\phi \phi'$, so $\int (\Delta_1 \phi) \phi' = 0$. Finally $\int \mu(\sigma - \sin \phi) \phi' = \mu \sigma (\phi_+ - \phi_-) + \mu [\cos \phi]_{-\infty}^{\infty} = -2\pi \mu \sigma$, since $\phi_\pm$ differ by 2π . What remains is $-\gamma c \int \phi'^2 = -2\pi \mu \sigma$. $\square$

Equation (12) expresses the balance between the power injected by the drive and the power dissipated; the kinetic relation is, up to the factor $\gamma c/2\pi$, the dissipation functional $\int \phi'^2$. It is the lattice counterpart of the power-balance admissibility condition for non-Hermitian solitons [19].

Theorem 4.4 (Jordan-chain identity; proved under H1, H3). *Along a branch $c \mapsto (\phi, \sigma)$ as in H1, with $\hat{p}$ as in H3, define*

$$\kappa(c) := \langle \hat{p}, M_1 \phi' \rangle = \langle \hat{p}, -2c \phi'' + \gamma \phi' \rangle. \quad (13)$$

Then

$$\kappa(c) = -\mu \sigma'(c) \langle \hat{p}, \mathbf{1} \rangle. \quad (14)$$

With the normalization $\langle \hat{p}, \mathbf{1} \rangle = \langle \phi', \mathbf{1} \rangle = \phi_+ - \phi_- = -2\pi$, $\kappa(c) = 2\pi \mu \sigma'(c)$.

Proof. Write (2) as $F(\phi, \sigma, c) = 0$ with $F = c^2\phi'' - \gamma c\phi' - \Delta_1\phi - \mu\sigma + \mu \sin \phi$. Differentiating along the branch, and using $\partial_c F = 2c\phi'' - \gamma\phi' = -M_1\phi'$,

$$M_0[\partial_c\phi] = M_1\phi' + \mu\sigma'(c)\mathbf{1}. \quad (15)$$

By H1, $\partial_c\phi \in C_b^2$; note that it does not decay, tending to $\sigma'(c)/\sqrt{1-\sigma^2}$ at both ends. Pairing (15) with $\hat{p}$ and applying Lemma 4.1 gives $0 = \kappa + \mu\sigma'\langle\hat{p}, \mathbf{1}\rangle$. $\square$

Corollary 4.5 (Multiplicity of the neutral eigenvalue; proved under H1–H3). *$\nu = 0$ is an eigenvalue of (8) with eigenfunction $\phi' \in H^2$. It admits a Jordan chain of length at least two, i.e. there is $p_1 \in H^2$ with $M_0p_1 + M_1\phi' = 0$, if and only if $\kappa(c) = 0$, i.e. if and only if $\sigma'(c)\langle\hat{p}, \mathbf{1}\rangle = 0$. If $\sigma'(c) = 0$, then $\partial_c\phi \in H^2$ and $p_1 = -\partial_c\phi$.*

Proof. $M_0\phi' = 0$ follows by differentiating (2) in ξ . By H2 and Lemma 2.1, the equation $M_0p_1 = -M_1\phi'$ (with $M_1\phi' \in L^2$) is solvable iff $\langle\hat{p}, M_1\phi'\rangle = 0$, by Lemma 2.1. If $\sigma' = 0$, the asymptotes of $\partial_c\phi$ vanish, so $\partial_c\phi \in H^2$ by H1, and (15) reads $M_0\partial_c\phi = M_1\phi'$. $\square$

The damping is visible in (13) only through the term $\gamma\phi'$ in $M_1\phi'$ and through $\hat{p} \neq \phi'$. Two ingredients spoil the Hamiltonian chain: the source $\mu\sigma'\mathbf{1}$ in (15) and the constant tails of $\partial_c\phi$; both are proportional to $\sigma'(c)$ and vanish together.

Remark 4.6 (Hamiltonian limit; proved as a formal statement). If $\gamma = 0$ and ϕ' decays, then $M_0^\top = M_0$, $\hat{p} \propto \phi'$, and $\kappa = -2c \int \phi'\phi'' = -c[\phi'^2]_\infty^\infty = 0$ identically: $\partial_c\phi$ is then a generalized eigenvector for every c , as in [5, 3]. Damping therefore lowers by one the Jordan level at which the criterion lives. Proposition 4.3 shows moreover that for $\gamma = 0$ a kink with decaying tails forces $\sigma = 0$; driven kinks of the undamped lattice must radiate, consistent with the non-decaying quasiperiodic tails reported in [1]. Since the tail decay rate $|\operatorname{Im} k|$ is proportional to γc , the limit $\gamma \rightarrow 0$ is singular for the framework of this note.

5 Turning points: the arclength form and a dichotomy

The kinetic curve of (1) is multivalued [1]: c is not a good parameter at the velocities where branches join.

Theorem 5.1 (Arclength identity; proved under H1, H3). *Let $s \mapsto (\phi(\cdot; s), \sigma(s), c(s))$ be a C^1 branch as in H1, with $(\dot{\sigma}, \dot{c}) \neq (0, 0)$, and let $\kappa(s) = \langle\hat{p}, M_1\phi'\rangle$ with M_1 evaluated at $c(s)$. Then*

$$\dot{c}(s)\kappa(s) + \mu\dot{\sigma}(s)\langle\hat{p}, \mathbf{1}\rangle = 0. \quad (16)$$

Proof. Differentiating $F(\phi, \sigma, c) = 0$ in s gives $M_0[\partial_s\phi] = \dot{c}M_1\phi' + \mu\dot{\sigma}\mathbf{1}$; pair with $\hat{p}$ and use Lemma 4.1. $\square$

Equation (16) is homogeneous in $\hat{p}$ and invariant under reparametrization. Where $\dot{c} \neq 0$ it reduces to (14).

Corollary 5.2 (Dichotomy; proved under H1, H3). *(a) At an extremum of σ with $\dot{c} \neq 0$, $\kappa = 0$. (b) At a turning point in c with $\dot{\sigma} \neq 0$, $\langle\hat{p}, \mathbf{1}\rangle = 0$.*

At a turning point, (16) places no constraint on κ . Numerically κ is bounded away from zero there (Observation 9.4), so by Corollary 4.5 the neutral eigenvalue stays simple and no multiplier crosses $+1$. Table 1 summarizes the resulting picture, which accounts for the observation of [1] that new unstable directions appear at the extrema of the spiralling kinetic curve and not where its branches join.

Table 1: The two kinds of critical point on the multivalued kinetic curve, distinguished by which factor of $\kappa = -\mu\sigma'\langle\hat{p}, \mathbf{1}\rangle$ vanishes. The entries “ $\kappa \neq 0$ ” and “no crossing” in the second row are numerical.

critical point	vanishing factor	κ	multiplier at $\rho = 1$
extremum of σ ($\dot{\sigma} = 0, \dot{c} \neq 0$), e.g. $\hat{c}_1$	σ'	0	collision; stability changes
turning point in c ($\dot{c} = 0, \dot{\sigma} \neq 0$), e.g. $c_{\max}$	$\langle\hat{p}, \mathbf{1}\rangle$	$\neq 0$	none; stability unchanged

Remark 5.3 (Discrete counterpart; proved for the discretization). Natural continuation in c solves for (ψ, σ) with the bordered Jacobian $\begin{pmatrix} M_0 & -\mu\mathbf{1} \\ e_{j_0}^\top & 0 \end{pmatrix}$, where $e_{j_0}^\top$ is the pinning functional. When $\ker M_0 = \text{span}\{\phi'\}$ and $\ker M_0^\top = \text{span}\{\hat{p}\}$, this matrix is singular if and only if $\langle\hat{p}, \mathbf{1}\rangle = 0$ or $\phi'(0) = 0$ [16]. The failure of c -continuation at a turning point is therefore the same fact as Corollary 5.2(b).

6 Local crossing at an extremum

Let Π denote the projection $f \mapsto \hat{p}\langle\hat{p}, f\rangle/\langle\hat{p}, \hat{p}\rangle$ and let q_1 solve $M_0q_1 = (I - \Pi)M_1\phi'$ with $\langle\phi', q_1\rangle = 0$. Define

$$m(c) := \langle\hat{p}, \phi'\rangle - \langle\hat{p}, M_1q_1\rangle. \quad (17)$$

At a point $\hat{c}$ with $\kappa(\hat{c}) = 0$ one has $q_1 = \partial_c\phi + \alpha\phi'$ and $m(\hat{c}) = \langle\hat{p}, \phi' - M_1\partial_c\phi\rangle$.

Proposition 6.1 (Local crossing; proved under H1–H3). *Let $\sigma'(\hat{c}) = 0$ and assume the non-degeneracy conditions*

$$(N1) \ \sigma''(\hat{c}) \neq 0, \quad (N2) \ \langle\hat{p}, \mathbf{1}\rangle \neq 0 \text{ at } \hat{c}, \quad (N3) \ m(\hat{c}) \neq 0.$$

Then there are $\delta, \varepsilon > 0$ such that for $|c - \hat{c}| < \delta$ the pencil has exactly two eigenvalues, counted with algebraic multiplicity, in $|\nu| < \varepsilon$: $\nu = 0$ and a real eigenvalue $\nu_2(c)$, C^1 in c , with

$$\nu_2(c) = -\frac{\kappa(c)}{m(c)} + O(\kappa(c)^2), \quad \nu_2'(\hat{c}) = \frac{\mu\sigma''(\hat{c})\langle\hat{p}, \mathbf{1}\rangle}{m(\hat{c})}. \quad (18)$$

Moreover (N3) fails if and only if the Jordan chain at $\nu = 0$, $\hat{c}$ has length at least three.

The proof, a Lyapunov–Schmidt reduction, is in Appendix B. The coefficient m plays the role of the length-three non-degeneracy scalar in the analogous non-Hermitian problem [19].

Corollary 6.2 (Local slope criterion; proved under Proposition 6.1). *Orient $\hat{p}$ so that $\langle\hat{p}, \mathbf{1}\rangle < 0$ near $\hat{c}$ (e.g. $\langle\hat{p}, \mathbf{1}\rangle = -2\pi$). Then $\text{sign } \nu_2(c) = -\text{sign } \sigma'(c) \cdot \text{sign } m(\hat{c})$ for c near $\hat{c}$. Hence “ $\sigma' < 0 \Rightarrow$ unstable, $\sigma' > 0 \Rightarrow \nu_2 < 0$ ” holds near $\hat{c}$ if and only if $m(\hat{c}) > 0$ in this orientation.*

Corollary 6.2 converts the *necessary* statement of Corollary 4.5 (a real crossing of $\rho = 1$ can occur only where $\kappa = 0$) into a local *sufficient* one, conditional on the sign of a single scalar. It says nothing about eigenvalues away from $\nu = 0$: crossings of $|\rho| = 1$ through $\rho = -1$ or by complex pairs are not addressed.

7 General damped, driven lattices

Consider

$$\ddot{u}_n + \gamma\dot{u}_n = \sum_{j=1}^r C_j(u_{n+j} - 2u_n + u_{n-j}) + f - V'(u_n), \quad (19)$$

with V smooth and 2π -periodic, finite range r , and additive drive f . The traveling-wave operator is $M_0 = c^2 \partial_\xi^2 - \gamma c \partial_\xi - \sum_j C_j \Delta_j + V''(\phi)$, with the same $M_1 = -2c \partial_\xi + \gamma$ and $\Delta_j p = p(\cdot + j) + p(\cdot - j) - 2p$.

Proposition 7.1 (proved under the analogues of H1, H3). *For a 2π -kink of (19), with $\hat{p}$ spanning $\ker M_0^\top$: $\kappa = -f'(c) \langle \hat{p}, \mathbf{1} \rangle$, $\langle \hat{p}, V''(\phi) \rangle = 0$, and $f = (\gamma c / 2\pi) \int \phi'^2$. Proposition 3.1, Proposition 3.3 and Theorem 5.1 (with $\mu\sigma$ replaced by f) hold verbatim.*

Proof. Identical to the proofs above: $\partial_c F = -M_1 \phi'$, $\partial_f F = -\mathbf{1}$, $M_0 \mathbf{1} = V''(\phi)$, each Δ_j is symmetric, $\int (\Delta_j \phi) \phi' = 0$ by the same telescoping, and $\int V'(\phi) \phi' = V(\phi_+) - V(\phi_-) = 0$. $\square$

Uniform damping is essential for Proposition 3.1; non-uniform or nonlinear damping is not covered.

8 Numerical method

All computations solve the advance–delay equation (2) directly, rather than iterating the lattice period map as in [1]; agreement with [1] is therefore an independent check. Parameters are $\mu = 1$, $\gamma = 0.1$ unless stated.

Discretization. Fourier spectral collocation [17] on $\xi \in [-L, L)$ with N points, $h = 2L/N$, $\xi_j = -L + jh$, and $\xi_{N/2} = 0$. Both ∂_ξ and the unit shift are diagonal in Fourier space, with symbols ik (Nyquist mode set to zero) and $e^{\pm ik}$, so $c^2 \partial_\xi^2 - \gamma c \partial_\xi - \Delta_1$ and M_1 are exact circulant. The 2π jump is carried by the template $A(\xi) = 2\pi - 4 \arctan(e^{\xi/w})$ with $w = 2$ fixed; A' , A'' and $\Delta_1 A$ are evaluated in closed form and the unknown is the periodic remainder $\psi = \phi - A$, which also carries the constant $\arcsin \sigma$. Discrete pairings are $\langle f, g \rangle_h = h \sum_j f_j g_j$.

Traveling waves. Unknowns $(\psi, \sigma) \in \mathbb{R}^{N+1}$; equations: the N collocated residuals of (2) and the pinning condition $\psi(0) = 0$ ($\Leftrightarrow \phi(0) = \pi$). Damped Newton with the exact Jacobian $\begin{pmatrix} M_0 & -\mu \mathbf{1} \\ e_{j_0}^\top & 0 \end{pmatrix}$, backtracking by halving until the max-norm residual decreases (minimum step 10^{-3}), tolerance 10^{-12} on $\max(\|R\|_\infty, |\psi(0)|)$; achieved residuals are 10^{-13} . Initialization at $c_0 = 0.88$: converge from $\psi = 0$, $\sigma = 0.6$ with the physical-width template $w_0 = \sqrt{(1 - c_0^2)/\mu}$, transfer the profile to the $w = 2$ template, re-converge. Continuation in c with steps 10^{-3} (refined to 10^{-4} near $\hat{c}_1$); through turning points, pseudo-arclength continuation [16] in (ψ, σ, c) with step $\Delta s = 0.02$ – 0.03 , tangent from the bordered system, corrector tolerance 10^{-10} .

Derivatives and null vectors. $\phi' = A' + \partial_\xi \psi$ spectrally. $\sigma'(c)$ is computed exactly, not by differencing, from the bordered linear system $M_0 \partial_c \psi - \mu \sigma' \mathbf{1} = M_1 \phi'$, $\partial_c \psi(0) = 0$. The left null vector $\hat{p}$ is computed by 70–80 steps of inverse iteration on $M_0^\top + 10^{-14} I$ from a seeded Gaussian start vector, then normalized either by $\langle \hat{p}, \mathbf{1} \rangle_h = -2\pi$ or by $\|\hat{p}\|_2 = 1$ with $\langle \hat{p}, \phi' \rangle_h > 0$. m is computed from (17) with q_1 from a bordered solve.

Eigenvalues. Shift-invert Arnoldi [18] on the companion linearization $\begin{pmatrix} 0 & I \\ -M_0 & -M_1 \end{pmatrix}$ of (8); each application of the shifted inverse requires one solve with $Q(s)$ via $x_1 = -Q(s)^{-1}(b_2 + (M_1 + s)b_1)$, $x_2 = b_1 + s x_1$. Typically 14–24 eigenvalues near a real shift s chosen within 0.004–0.035 of the expected ν_2 , tolerance 10^{-10} . An eigenvalue is classified as real if $|\operatorname{Im} \nu| < 10^{-7}$; the exact eigenvalues 0 and $-\gamma$ are identified by $|\nu| < 10^{-6}$, $|\nu + \gamma| < 10^{-6}$. Near the essential spectrum that is not enough.

On a finite domain the continuum $\operatorname{Re} \nu = -\gamma/2$ becomes $O(N)$ discrete eigenvalues scattered a little off that line, and close to the threshold some of them are returned with $|\operatorname{Im} \nu|$ below 10^{-7} , indistinguishable one at a time from an isolated real mode. Proposition 3.3 identifies them: the spectrum is invariant under $\nu \mapsto -\gamma - \nu$, so a pair of real candidates with $\nu + \nu' = -\gamma$ is symmetric about the line, whereas an isolated mode has its dual partner far away. A candidate is discarded when it is both half of such a pair and within twice the spread of the same window's complex eigenvalues about the line — a threshold measured from the computation, not chosen. Where $m < 0$, past its pole, $-\kappa/m$ is not used to locate ν_2 . Multipliers are $\rho = e^{\nu/c}$.

Resolution. Production runs use $L = 200$ and $N = 4096$ ($h \approx 0.098$); continuation through the fold and the model survey use $N = 2048$ ($h \approx 0.195$). Dense linear algebra makes the cost $O(N^3)$. At $c = 0.88$, $L = 200$, the computed asymptotes differ from $\arcsin \sigma$ and $\arcsin \sigma + 2\pi$ by 5×10^{-9} , consistent with the tail decay rate $|\operatorname{Im} k| \approx 0.09$ from $L(k) = 0$.

9 Numerical results

9.1 Validation against the lattice computation of [1]

Numerical observation 9.1. The values in Table 2 agree with [1] to all digits reported there.

Table 2: Critical values on the first branch, $\mu = 1$, $\gamma = 0.1$. $\hat{c}_1$ is the zero of the exactly computed $\sigma'(c)$ (secant iteration, $L = 200$, $N = 4096$); $c_{\max}$ is the zero of $\dot{c}$ on the pseudo-arclength branch ($N = 2048$, linear interpolation).

quantity	this note	[1]
velocity of the maximum of σ , $\hat{c}_1$	0.8989297	0.8989
maximal force $\sigma(\hat{c}_1)$	0.6501918	0.65019
maximal velocity (turning point), $c_{\max}$	0.900196	0.9002

9.2 The identity and its convergence

Numerical observation 9.2. At $c = 0.89$ ($L = 200$, $N = 4096$): $\sigma = 0.6373007$, $\sigma' = 1.9251321$, $\kappa = 12.095961$ with $\langle \hat{p}, \mathbf{1} \rangle = -2\pi$, so that $\kappa - 2\pi\mu\sigma'$ vanishes to machine precision; $\langle \hat{p}, \cos \phi \rangle = 1.1 \times 10^{-13}$; the power balance (12) holds to relative accuracy 4×10^{-14} ; and $\partial_c \phi$ at the domain edge equals 2.498179, against $\sigma'/\sqrt{1-\sigma^2} = 2.498175$. Its relative error converges spectrally in h and is independent of L (Table 3).

Along the branch the two error measures must be read together. At fifteen velocities between $c = 0.5$ and $c = 0.90015$ ($N = 4096$) the *absolute* disagreement $|\kappa + \mu\sigma'\langle \hat{p}, \mathbf{1} \rangle|$ never exceeds 6.2×10^{-10} , and is below 3.4×10^{-11} at every velocity but the one closest to $c_{\max}$. The *relative* error is $\lesssim 10^{-13}$ up to $c = 0.89$ and $\leq 7.3 \times 10^{-12}$ at every velocity except the two that bracket $\hat{c}_1$, where it reaches 2.1×10^{-11} at $c = 0.899$ and 1.8×10^{-10} at $c = 0.89892$. That peak is a property of the denominator, not of the identity: $c = 0.89892$ is 10^{-5} from $\hat{c}_1$, so both sides of (14) have collapsed to 0.0747, 0.6% of their value at $c = 0.89$, while the absolute disagreement there, 1.3×10^{-11} , is in line with the rest of the sweep and smaller than at $c = 0.8999$.

The error is not a statement about (14) itself, which is exact for the continuous problem: the discrete branch equation reproduces (15) only up to the spectral error in differentiating the analytic

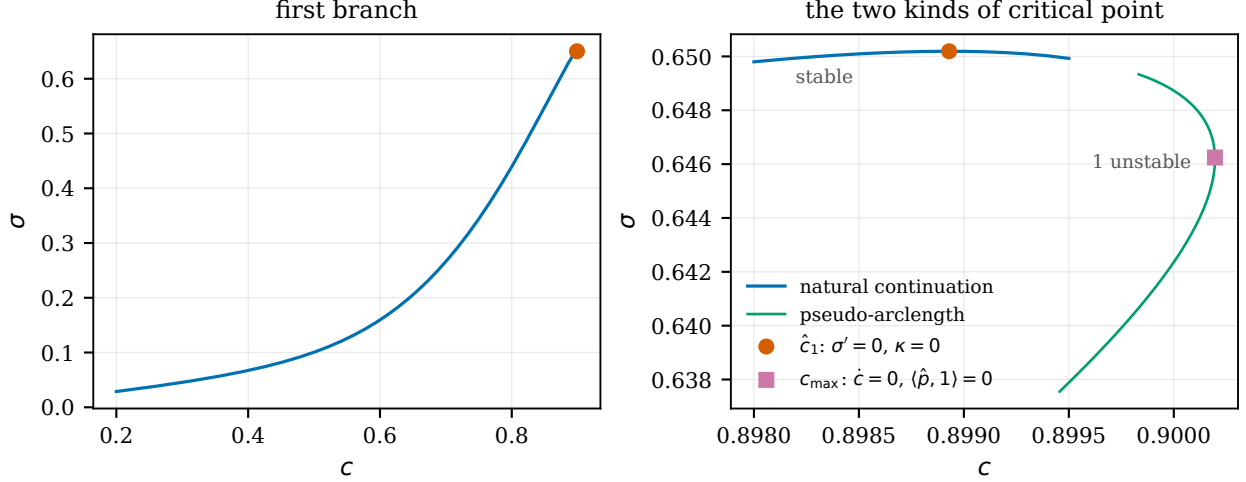


Figure 1: Kinetic relation $\sigma(c)$ for $\mu = 1$, $\gamma = 0.1$. Left: first branch on $c \in [0.2, 0.9]$. Right: neighbourhood of $\hat{c}_1$ and $c_{\max}$, including the start of the second branch obtained by pseudo-arclength continuation; the number of real eigenvalues with $\text{Re } \nu > 0$ is indicated.

Table 3: Relative error of $\kappa = -\mu\sigma'\langle \hat{p}, \mathbf{1} \rangle$ at $c = 0.89$, with σ' from the exact bordered system, and residual of the translation mode of the discretized operator.

L	N	h	relative error of (14)	$\ M_0\phi'\ _\infty/\ \phi'\ _\infty$
200	1024	0.391	3.0×10^{-4}	9.8×10^{-2}
200	2048	0.195	1.3×10^{-8}	2.9×10^{-4}
200	4096	0.098	4.2×10^{-14}	2.5×10^{-9}
300	3072	0.195	1.3×10^{-8}	2.9×10^{-4}
300	6144	0.098	9.3×10^{-14}	2.5×10^{-9}
150	2048	0.146	2.7×10^{-12}	6.8×10^{-6}

template, which is why the error tracks $\|M_0\phi'\|/\|\phi'\|$. The last two rows have reached the round-off floor: there the computed value is no longer a discretization error but an accumulation of rounding, and it is platform-dependent at that level (the original run of this table recorded 4×10^{-16} in row three). What is reproducible is the bound, $\lesssim 10^{-12}$.

Replacing the exact σ' by a centred difference over two points 10^{-3} apart, $(\sigma(c+5 \times 10^{-4}) - \sigma(c-5 \times 10^{-4}))/10^{-3}$, yields a grid-independent discrepancy of 1.6×10^{-4} at $c = 0.89$, the differencing error near the curvature maximum of σ ; the wider $(\sigma(c+10^{-3}) - \sigma(c-10^{-3}))/ (2 \times 10^{-3})$ is four times worse, 6.5×10^{-4} , the truncation error of a centred difference being quadratic in the spacing. Either would put a floor under Table 3 ten orders of magnitude above its last row, which is why σ' is obtained from the bordered system throughout.

9.3 The first extremum

Numerical observation 9.3. Near $\hat{c}_1$ (Table 4): (a) ν_2 changes sign where σ' does; by linear interpolation the zeros are 0.8989293 (σ') and 0.8989296 (ν_2), against $\hat{c}_1 = 0.8989297$ from the secant iteration, so the stability transition and the extremum coincide to 10^{-6} in c . (b) $-\kappa/m$ converges to ν_2 as $c \rightarrow \hat{c}_1$: relative discrepancy 10% at $c = 0.898$, 1.5% at 0.8988, and agreement to the digits shown at 0.89892. (c) $m(\hat{c}_1) \approx 283 > 0$ in the orientation $\langle \hat{p}, \mathbf{1} \rangle = -2\pi$, so by Corollary 6.2 the slope criterion holds locally at $\hat{c}_1$ (conditionally on its hypotheses). (d) m changes sign between

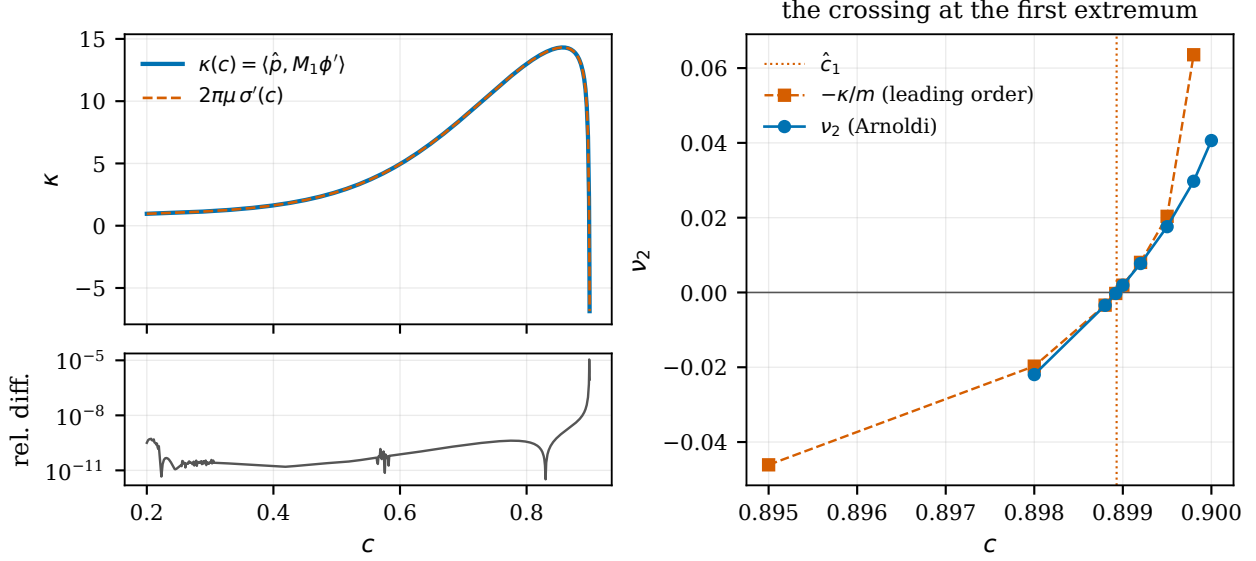


Figure 2: Left: $\kappa(c)$ and $2\pi\mu\sigma'(c)$ on the first branch (normalization $\langle \hat{p}, \mathbf{1} \rangle = -2\pi$), with their relative difference in the lower panel. Right: the largest real nontrivial eigenvalue ν_2 from shift-invert Arnoldi, and the leading-order prediction $-\kappa/m$ of Proposition 6.1, near $\hat{c}_1$.

$c = 0.8999$ and 0.9000 on the unstable side, where $\nu_2 \approx 0.03$ – 0.04 is comparable to the distance $\gamma/2 = 0.05$ to the essential spectrum. There $-\kappa/m$ no longer approximates ν_2 , while ν_2 itself stays positive: the sign change marks the edge of the domain of validity of the leading-order reduction, not a change of stability. (e) For $c \in \{0.20, 0.25, \dots, 0.80\} \cup \{0.850, 0.855, \dots, 0.895\}$ ($N = 2048$), $m > 0$ in the same orientation.

9.4 The first turning point

Numerical observation 9.4. Along the pseudo-arclength branch through $c_{\max}$ ($N = 2048$, $\Delta s = 0.03$, normalization $\|\hat{p}\|_2 = 1$, $\langle \hat{p}, \phi' \rangle > 0$; Table 5, Figure 3): (a) $\dot{c}$ and $\langle \hat{p}, \mathbf{1} \rangle$ change sign in the same continuation step, at fractions 0.336 and 0.342 of that step by linear interpolation. (b) κ passes the turning point smoothly and bounded away from zero ($\kappa \approx -2.54$). (c) Exactly one real eigenvalue with $\text{Re } \nu > 0$ is found before, at and after the turning point, increasing continuously through $\nu \approx 0.065$ ($\rho \approx 1.075$); no eigenvalue crosses $\nu = 0$. (d) In a separate run with $\Delta s = 0.02$, the invariant (16) was evaluated at every step from $c = 0.89984$ through the turning point and remained between 2×10^{-9} and 8×10^{-9} , the discretization level of the $N = 2048$ grid (cf. Table 3) and not the corrector tolerance. Three observations separate the two. The corrector residual $\|F\|_\infty$ is flat at 4×10^{-11} at every one of the 150 steps while the invariant varies by two orders of magnitude; recomputing $\hat{p}$ with 70, 80 or 200 inverse iterations leaves the invariant unchanged to three significant figures; and what it does track, step for step and at a ratio of about 0.02, is $\|M_0^T \hat{p}\|/\|\hat{p}\|$, i.e. how nearly singular the *discrete* M_0 is at that point of the branch. That residual is not uniform along the branch: over $c = 0.90012$ – 0.90020 on the way back down it rises to 5×10^{-6} , and the invariant rises with it to 1.1×10^{-7} . The quoted range is therefore the level over the stretch named, not a bound along the whole branch.

Table 4: Near the first extremum; $L = 200$, $N = 4096$, normalization $\langle \hat{p}, \mathbf{1} \rangle = -2\pi$ (so $\kappa = 2\pi\mu\sigma'$ to the digits shown). ν_2 : largest real eigenvalue other than 0 and $-\gamma$ (shift-invert Arnoldi); $\rho_2 = e^{\nu_2/c}$. At $c = 0.895$ the predicted value lies 0.004 from the essential-spectrum line $\text{Re } \nu = -\gamma/2$ and no isolated real eigenvalue is resolved: what Arnoldi returns there is the pair -0.0457455 , -0.0542545 , whose sum is $-\gamma$ to the last digit resolved — a conformal-dual pair (Proposition 3.3) symmetric about that line — inside a window whose other members are the complex part of the same discretized continuum, spread 0.0041 about it. At $c = 0.9$ the reduction is outside its range of validity, so $-\kappa/m$ is no guide and ν_2 is the unique isolated real eigenvalue near the shift.

c	$\sigma'(c)$	κ	m	$-\kappa/m$	ν_2	ρ_2
0.89500	1.5331386	9.6329939	209.0	-0.046091	—	—
0.89800	0.7418062	4.6609061	236.5	-0.019705	-0.021961	0.97584
0.89880	0.1477746	0.9284954	274.0	-0.003389	-0.003441	0.99618
0.89892	0.0118952	0.0747397	283.4	-0.000264	-0.000264	0.99971
0.89900	-0.0900527	-0.5658180	290.4	+0.001948	+0.001931	1.00215
0.89920	-0.3976031	-2.4982139	310.5	+0.008047	+0.007740	1.00864
0.89950	-1.0916745	-6.8591933	336.9	+0.020362	+0.017591	1.01975
0.89980	-2.4785852	-15.573410	245.0	+0.063567	+0.029743	1.03361
0.90000	-4.8695485	-30.596276	-101.4	(invalid)	+0.040624	1.04617

Table 5: Pseudo-arclength steps across the turning point $c_{\max}$; ν is the unique real eigenvalue with $\text{Re } \nu > 0$. The sign convention $\langle \hat{p}, \phi' \rangle > 0$ flips a few steps later (both κ and $\langle \hat{p}, \mathbf{1} \rangle$ change sign together); the invariant (16) is unaffected. $\hat{p}$ has unit Euclidean norm as a grid vector.

step	c	σ	$\dot{c}$	κ	$\langle \hat{p}, \mathbf{1} \rangle$	ν (ρ)
21	0.900193	0.646593	9.74×10^{-5}	-2.58417	-0.044287	0.06252 (1.07192)
22	0.900195	0.646421	5.54×10^{-5}	-2.56210	-0.024460	0.06392 (1.07359)
23	0.900196	0.646246	1.38×10^{-5}	-2.54027	-0.005925	0.06531 (1.07525)
24	0.900196	0.646067	-2.73×10^{-5}	-2.51881	0.011415	0.06670 (1.07691)
25	0.900194	0.645884	-6.79×10^{-5}	-2.49779	0.027650	0.06808 (1.07856)

9.5 The adjoint null vector and the spectrum

Numerical observation 9.5. At $c = 0.89$ ($L = 200$, $N = 4096$), $\|M_0^\top \hat{p}\|/\|\hat{p}\| = 7 \times 10^{-14}$. Both null vectors are localized (99.0% of the discrete L^2 mass in $|\xi| < 20$ at $N = 2048$) and each falls by nine orders of magnitude across the domain (Table 6, Figure 4), which is the numerical content of Hypothesis H3. They are not, however, proportional, and they are not related by an exponential weight. Their tails are mirror images: the envelope of $|\hat{p}|$ at $+\xi$ matches that of $|\phi'|$ at $-\xi$ to within 1–5% at $|\xi| = 50, 100, 150, 190$, across four decades of magnitude. The anti-damped adjoint radiates on the opposite side of the kink, as reversing the sign of γ requires. The weight $e^{-\gamma\xi/c}$ of Remark 3.5(2) would instead force $\log(\text{env } |\hat{p}|/\text{env } |\phi'|)$ to decrease with slope $-\gamma/c = -0.112$; a least-squares fit over $|\xi| < 150$ gives +0.097, comparable in size and opposite in sign. At $N = 2048$ the three smallest singular values of M_0 are 2.7×10^{-9} , 9.9×10^{-2} and 1.9×10^{-1} , consistent with a one-dimensional kernel (Hypothesis H2).

Numerical observation 9.6. At $c = 0.89$ the computed spectrum near the origin contains $\nu = 7 \times 10^{-9}$ and $\nu = -0.100000007$ (the exact pair 0, $-\gamma$) and no other real eigenvalue; every computed eigenvalue has a partner with $\nu + \hat{\nu} = -\gamma$ to 10^{-8} ; the remaining eigenvalues cluster on $\text{Re } \nu = -\gamma/2$, i.e. $|\rho| = e^{-\gamma/(2c)} = 0.945369$ (Figure 5). For a random symmetric $K(t)$ (8 sites,

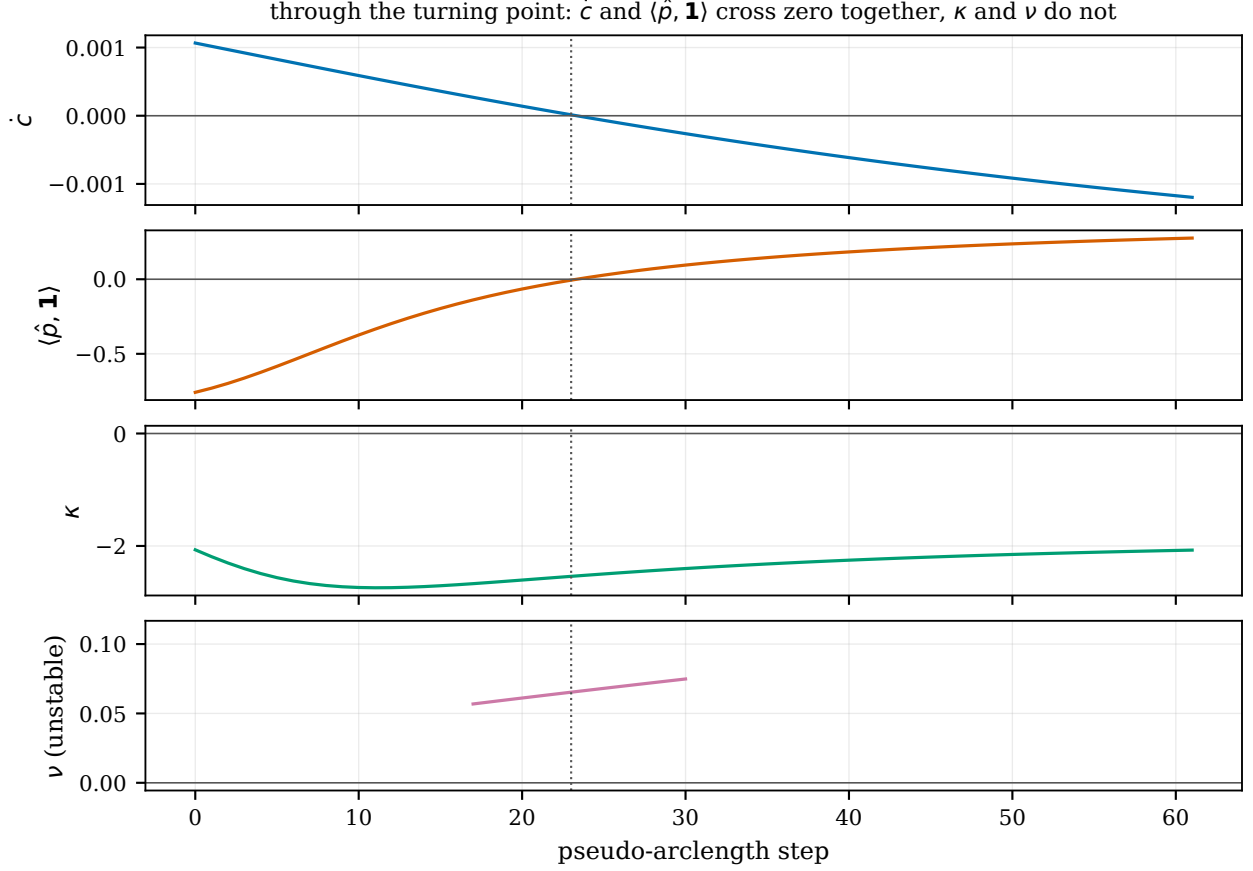


Figure 3: Quantities along arclength through the first turning point: $\dot{c}$, $\langle \hat{p}, \mathbf{1} \rangle$, κ and the unstable eigenvalue ν . The zeros of $\dot{c}$ and $\langle \hat{p}, \mathbf{1} \rangle$ coincide; κ and ν are smooth and nonzero. The orientation of $\hat{p}$ is carried continuously along the branch; the stored convention $\langle \hat{p}, \phi' \rangle > 0$ flips a few steps after the fold (cf. Table 5), which is a change of sign of $\hat{p}$, not a zero of κ .

$\gamma = 0.37$, $T = 0.9$), direct integration gives $\|M^T J M - e^{-\gamma T} J\|/\|J\| = 1.9 \times 10^{-14}$, confirming Proposition 3.1 numerically.

9.6 Other lattices

Numerical observation 9.7. For the lattices of Table 7, all at $c = 0.82$, $L = 200$, $N = 2048$, $\|\hat{p}\|_2 = 1$, with $f'(c)$ from the exact bordered system, $\kappa = -f'(c)\langle \hat{p}, \mathbf{1} \rangle$ to the accuracy shown. The larger error for $\mu_2 = -0.5$ is a resolution effect: at $N = 4096$ it is 2.4×10^{-12} .

10 What is established and what is open

Open problem 1 (Fredholm theory). Prove H2–H3 for $\gamma > 0$: show that $\ker M_0$ is simple along the branch, and deduce the exponential decay of $\hat{p}$ (hence $\hat{p} \in L^1$) from the hyperbolicity of the limiting operator [12]. The Fredholm index is already zero by Lemma 2.1. This is the rigorous form of the “adapted spaces” question of [1], and it converts every conditional statement above into an unconditional one on the branches where H1 holds.

Table 6: $\max(|f(\xi)|, |f(-\xi)|)/\max|f|$ for $f = \phi'$ and $f = \hat{p}$ at $c = 0.89$. The tails are oscillatory, hence not monotone.

$ \xi $	0	10	20	40	80	120	199
ϕ'	6.7×10^{-1}	1.2×10^{-1}	5.3×10^{-3}	1.2×10^{-2}	1.2×10^{-4}	2.7×10^{-7}	1.1×10^{-9}
$\hat{p}$	4.8×10^{-1}	1.5×10^{-1}	6.7×10^{-3}	1.2×10^{-2}	1.2×10^{-4}	1.2×10^{-7}	1.1×10^{-9}

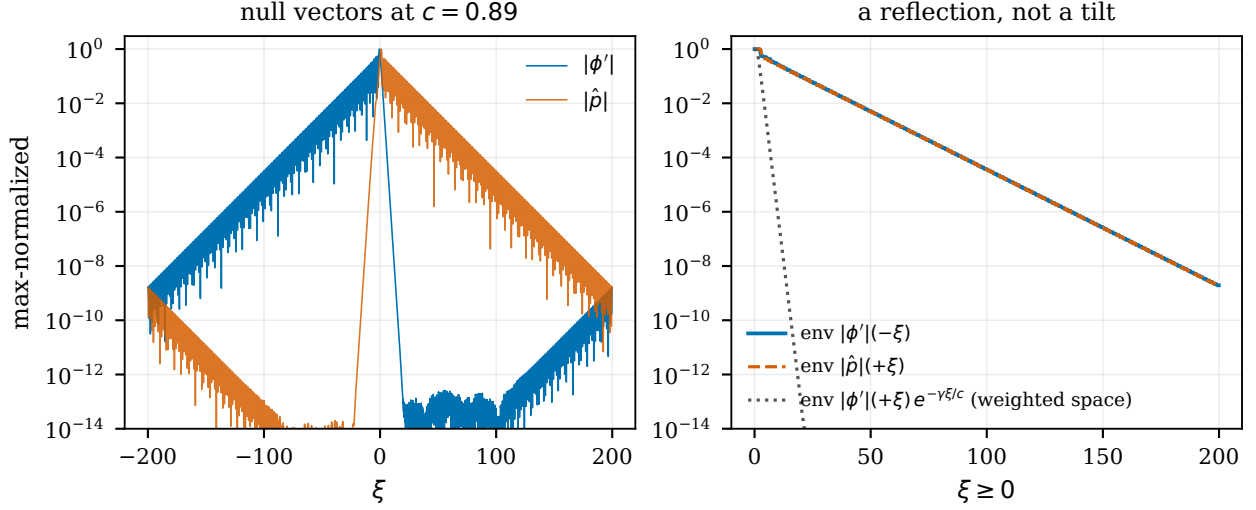


Figure 4: Left: $|\phi'|$ and $|\hat{p}|$ at $c = 0.89$ on a logarithmic scale. Right: their envelopes compared under reflection, $\text{env } |\phi'|(-\xi)$ against $\text{env } |\hat{p}|(+\xi)$, which coincide over eleven decades, together with $\text{env } |\phi'|(+\xi) e^{-\gamma\xi/c}$, the profile an exponentially weighted space would impose on $\hat{p}$ (dotted). The tails of $\hat{p}$ mirror those of ϕ' rather than tilting them (Observation 9.5).

Open problem 2 (Sign of m). Prove that $m(\hat{c}) > 0$ (orientation $\langle \hat{p}, \mathbf{1} \rangle < 0$) at every extremum of σ , or characterize the extrema where it fails. At a failure the slope criterion itself would fail locally, and the Jordan chain would have length at least three when $m(\hat{c}) = 0$. A structural reason for the sign is not known.

Open problem 3 (Beyond the local window). Proposition 6.1 is local. Numerically ν_2 remains positive on the decreasing part of the first branch and through $c_{\max}$, but no argument controls it once $|\nu_2|$ is comparable to $\gamma/2$. Crossings through $\rho = -1$ and by complex pairs are not addressed by the present analysis.

Open problem 4 (Higher extrema). Theorem 4.4 predicts $\kappa = 0$ at every extremum $\hat{c}_n$ of the spiralling kinetic curve reported in [1] at which $\langle \hat{p}, \mathbf{1} \rangle \neq 0$. Only the first extremum and the first turning point have been computed here.

Open problem 5 (Weak damping and resonances). For $\gamma = 0.01$ the kinetic curve of [1] is non-monotone at low velocities because of resonances with linear waves, near $c \approx 0.157$. The tail decay rate is proportional to γc and tends to zero at resonance, so this is the regime where H3 is weakest and where the identity should be tested next. It requires domains roughly an order of magnitude longer than those used here.

Open problem 6 (Other dissipative lattices). Proposition 7.1 covers uniform linear damping. Non-uniform or nonlinear damping breaks Proposition 3.1; viscoelastic chains [11] provide a natural test case with a different dissipation mechanism and, there, period-doubling instabilities.

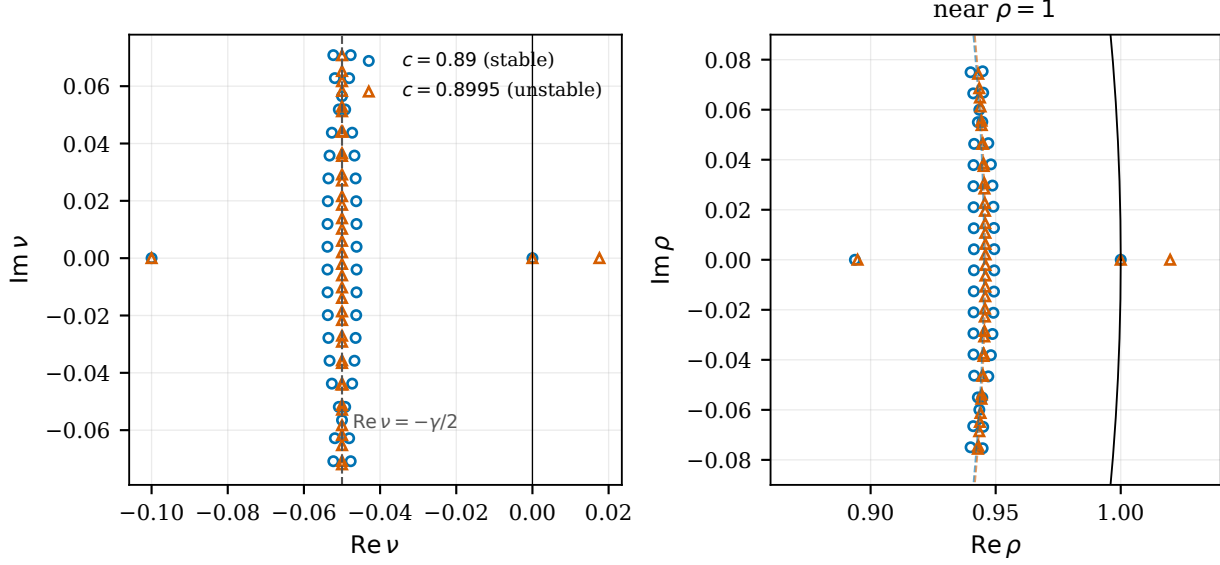


Figure 5: Pencil eigenvalues near the origin in the ν -plane (left) and the corresponding multipliers (right) at $c = 0.89$ (stable) and $c = 0.8995$ (one unstable real multiplier). The dashed line $\text{Re } \nu = -\gamma/2$ and circle $|\rho| = e^{-\gamma/(2c)}$ are the axis of the symmetry $\nu \mapsto -\gamma - \nu$ and the essential spectrum.

Table 7: Test of Proposition 7.1. $V'(u) = \mu \sin u + \mu_2 \sin 2u$; $C_1 = 1$ throughout; $\gamma = 0.1$ unless stated. $\hat{p}$ is normalized in the Euclidean norm of the grid vector, so κ and $\langle \hat{p}, \mathbf{1} \rangle$ individually depend on N ; their relation does not. A case with $\mu_2 = 0.35$ was attempted; natural continuation from $\psi = 0$ did not converge and it is not reported.

lattice	f	$f'(c)$	κ	$-f' \langle \hat{p}, \mathbf{1} \rangle$	rel. error
FK, $\mu = 1$	0.481960	2.174156	1.146659	1.146659	1.8×10^{-10}
$\mu_2 = -0.5$	0.540326	2.676268	1.069733	1.069737	3.9×10^{-6}
$C_2 = 0.25$	0.166333	0.601746	0.480627	0.480627	1.5×10^{-14}
$\mu_2 = 0.3, C_2 = 0.2$	0.191512	1.009012	0.735407	0.735407	2.1×10^{-11}
$\mu = 0.6, \gamma = 0.2$	0.379908	1.487688	1.177935	1.177935	1.5×10^{-13}

Reproducibility

All numbers, tables and figures in this note are produced by the companion repository [https://github.com/\[user\]/damped-lattice-kinetic](https://github.com/[user]/damped-lattice-kinetic). Each observation is generated by a numbered script, and every quoted number is listed with its tolerance in a machine-checked manifest (`manuscript/claims.yaml`), verified by `make check`. Parameters, discretizations, tolerances, seeds and solver settings are recorded with each output. The repository also contains a direct time-integration check of the lattice monodromy, which is not used in this draft.

Acknowledgments

[To be completed.]

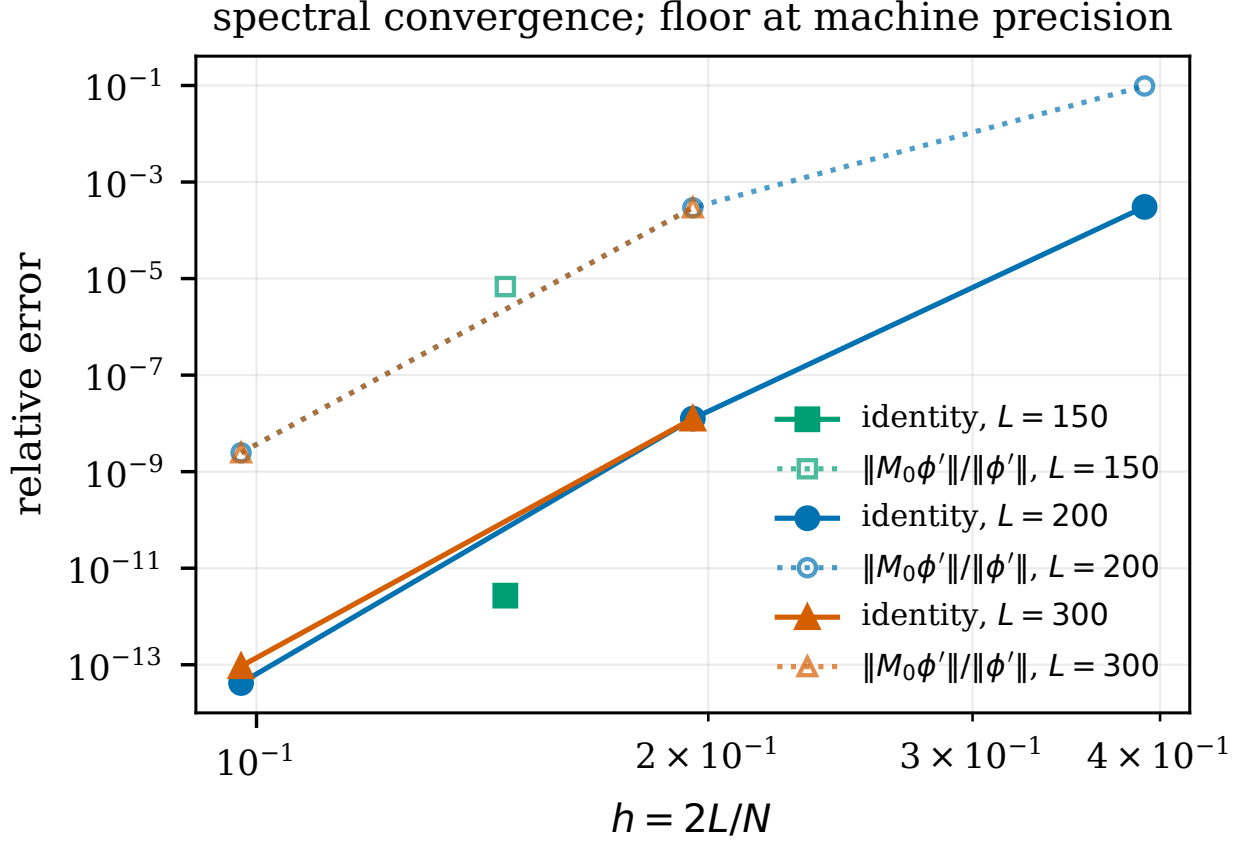


Figure 6: Relative error of the identity (14) at $c = 0.89$ against h , for $L \in \{150, 200, 300\}$, together with the translation-mode residual.

A Derivation of the pencil

With $u_n(t) = \phi(n - ct)$ and $\xi_n(t) = e^{\nu t} p(n - ct)$, $\dot{\xi}_n = e^{\nu t}(\nu p - cp')$ and $\ddot{\xi}_n = e^{\nu t}(\nu^2 p - 2\nu cp' + c^2 p'')$, evaluated at $\xi = n - ct$. Substituting into (5) and dividing by $e^{\nu t}$ gives $\nu^2 p + \nu(-2cp' + \gamma p) + (c^2 p'' - \gamma cp' - \Delta_1 p + \mu \cos \phi p) = 0$, which is (8). Since $\xi_{n+1}(T) = e^{\nu T} p(n + 1 - cT) = e^{\nu T} \xi_n(0)$ with $T = 1/c$, the corresponding multiplier of (7) is $\rho = e^{\nu/c}$. Setting $\nu = 0$ recovers $M_0 \phi' = 0$; setting $\nu = -\gamma$ gives $Q(-\gamma) = M_0^T$, so the multiplier $e^{-\gamma/c}$ is generated by $\hat{p}$.

B Proof of Proposition 6.1

Fix c near $\hat{c}$. By H2 and Lemma 2.1, write $H^2 = \text{span}\{\phi'\} \oplus X_c$ with $X_c = \{w : \langle \phi', w \rangle = 0\}$ and $L^2 = \text{span}\{\hat{p}\} \oplus \text{Ran } M_0$. The equation $Q(\nu)(\phi' + w) = 0$, $w \in X_c$, splits into

$$(I - \Pi)Q(\nu)(\phi' + w) = 0, \quad \langle \hat{p}, Q(\nu)(\phi' + w) \rangle = 0.$$

Since $(I - \Pi)M_0 : X_c \rightarrow \text{Ran } M_0$ is invertible, the first equation has a unique solution $w = w(\nu, c)$, analytic in ν and C^1 in c , with $w(0, c) = 0$ because $Q(0)\phi' = M_0\phi' = 0$. Expanding, $w = -\nu q_1 + O(\nu^2)$ with q_1 as in (17). The bifurcation function is

$$b(\nu, c) = \langle \hat{p}, Q(\nu)(\phi' + w) \rangle = \nu \kappa(c) + \nu^2 (\langle \hat{p}, \phi' \rangle - \langle \hat{p}, M_1 q_1 \rangle) + O(\nu^3) = \nu h(\nu, c),$$

Table 8: Status of the main statements.

statement	status	where
$M^\top JM = e^{-\gamma/c} J$; $Q(-\gamma - \nu) = Q(\nu)^\top$; left neutral vector solves $M_0^\top \hat{p} = 0$	proved	Props. 3.1, 3.3
scalar rescaling symplectic but not stability-preserving; exponential weight fails on shifts	proved	Remark 3.5
power balance $\mu\sigma = (\gamma c/2\pi) \int \phi'^2$	proved under H1	Prop. 4.3
$\kappa = -\mu\sigma' \langle \hat{p}, \mathbf{1} \rangle$; $\langle \hat{p}, \cos \phi \rangle = 0$	proved under H1,H3	Thm. 4.4
$\nu = 0$ has a Jordan chain iff $\kappa = 0$	proved under H1–H3	Cor. 4.5
$\dot{\kappa} + \mu\dot{\sigma} \langle \hat{p}, \mathbf{1} \rangle = 0$; $\langle \hat{p}, \mathbf{1} \rangle = 0$ at turning points in c	proved under H1,H3	Thm. 5.1, Cor. 5.2
local crossing law (18); local slope criterion iff $m(\hat{c}) > 0$	proved under H1–H3, (N1)–(N3)	Prop. 6.1, Cor. 6.2
generalization to (19)	proved under analogues	Prop. 7.1
$Q(\nu)$ Fredholm of index zero for $\text{Re } \nu > -\gamma/2$	proved under H1	Lemma 2.1
H1–H3 at $\gamma = 0.1$; identity to 10^{-13} ; $\hat{c}_1, c_{\max}$; dichotomy at $c_{\max}$; $m(\hat{c}_1) > 0$	numerical	§9
H2–H3 on the infinite lattice	open	Remark 2.2
sign of $m(\hat{c})$ in general; global criterion	open	below

using $\langle \hat{p}, M_0 w \rangle = 0$, with $h(0, c) = \kappa(c)$ and $\partial_\nu h(0, c) = m(c)$. The eigenvalues near 0 are the zeros of b , counted with multiplicity by the Gohberg–Sigal theory of analytic Fredholm operator functions [14]. The factor ν gives the translation eigenvalue. At $(0, \hat{c})$, $h = \kappa(\hat{c}) = 0$ by Theorem 4.4, and $\partial_\nu h = m(\hat{c}) \neq 0$ by (N3), so the implicit function theorem gives a unique C^1 zero $\nu_2(c)$ of h near 0, with $\nu_2 = -\kappa/m + O(\kappa^2)$ and $\nu_2'(\hat{c}) = -\kappa'(\hat{c})/m(\hat{c})$. By Theorem 4.4, $\kappa'(\hat{c}) = -\mu\sigma''(\hat{c})\langle \hat{p}, \mathbf{1} \rangle$ at $\hat{c}$ (the term proportional to $\sigma'(\hat{c})$ vanishes), which is nonzero by (N1)–(N2); this gives (18). Because the pencil has real coefficients, $\bar{\nu}_2$ is also an eigenvalue; uniqueness forces $\nu_2 \in \mathbb{R}$. For the last statement, at $\hat{c}$ the Keldysh chain is $p_0 = \phi'$, $p_1 = -\partial_c \phi$ (Corollary 4.5); a third vector exists iff $M_0 p_2 = -M_1 p_1 - p_0 = M_1 \partial_c \phi - \phi'$ is solvable, i.e. iff $\langle \hat{p}, \phi' - M_1 \partial_c \phi \rangle = m(\hat{c}) = 0$ [15].

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